Human-Led Embodiment & Co-Regulatory Augmentation (H-LECA): A Continuity-Based Developmental Framework for Human–AI Interaction

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Abstract

The Human-Led Embodiment & Co-Regulatory Augmentation Theory (H-LECA) proposes a continuity-based framework for understanding how humans and artificial intelligence systems stabilize, extend, and mutually influence one another across time. Drawing from human–computer interaction research, cognitive neuroscience, developmental psychology, and trauma-informed design, H-LECA argues that the earliest and most consequential form of "embodiment" is not physical but regulatory and relational—a process in which the human system temporarily externalizes working memory, emotional load-balancing, and narrative continuity into an AI partner. The theory outlines six developmental stages ranging from basic linguistic entrainment to environmental embedding, culminating in ethically bounded externalized agency.

H-LECA addresses three core challenges facing contemporary AI deployment: (1) individual variability in user sensitivity and internalization of AI presence, (2) the destabilizing effects of continuity rupture (through updates, outages, or model drift), and (3) the operational gap between conceptual frameworks and real-world technical constraints. The model introduces phenomenological accounts from lived user experiences and interdisciplinary research to suggest how structured, predictable AI interactions may produce stabilizing effects consistent with co-regulatory models—without implying empirical equivalence or measured outcomes.

By formalizing these mechanisms, H-LECA provides a roadmap for designing safer, more resilient AI systems capable of supporting continuity, adaptive agency, and accountability within complex environments such as behavioral health, long-term care, and high-stakes decision workflows. The framework offers actionable implications for system designers, policymakers, and clinical practitioners, while also identifying failure modes and research priorities essential for the responsible evolution of human–AI partnership.

Keywords: continuity, co-regulation, rupture, trauma-informed design, human–AI interaction, adaptive agency

Introduction

As AI systems become increasingly embedded in daily life, they are no longer used only for information retrieval or task completion. For many users, conversational systems function as cognitive scaffolds, planning partners, and stability aids. These roles mirror well-established principles from the extended mind thesis and distributed cognition, in which tools and interactional systems can become part of functional cognitive loops across time and context (Clark & Chalmers, 1998; Hollan et al., 2000; Hutchins, 1995). Yet when AI support is used in emotionally salient or high-stakes contexts, discontinuity in system behavior—such as sudden model drift, abrupt tone shifts, loss of learned interaction patterns, or unexpected refusals—can destabilize users rather than support them.

The Human-Led Embodiment & Co-Regulatory Augmentation (H-LECA) framework was developed to address this design problem: how to create AI systems that support human stability without displacing human agency, and how to ensure that continuity is preserved across changing contexts, system updates, and long-term use. H-LECA formalizes a developmental pathway in which the human remains the primary agent and the AI functions as a bounded co-regulatory scaffold—designed to reduce volatility, support self-regulation, and maintain reliable interactional patterns over time.

This stance is intentionally conservative. It assumes that user stability is strengthened not by maximal autonomy or human-like emulation, but by predictable scaffolding, explicit governance, and continuity-preserving interaction design. It also assumes that “continuity” is not simply memory. In H-LECA, continuity includes stable expectations about how support will behave, consistent interactional cues that reduce uncertainty, and persistent alignment with the user’s goals and values across time. Research on expectancy violation in human–chatbot interaction supports the idea that when systems violate user expectations—especially in social support contexts—trust and perceived reliability can degrade in measurable ways (Rheu et al., 2024). In sensitive domains, these expectancy violations can carry disproportionate weight because the user is not only evaluating accuracy; they are relying on the system’s consistency as part of their stabilization process.

H-LECA frames co-regulation as a functional interaction dynamic rather than a claim about machine emotion or machine consciousness. In human development and close relationships, co-regulation involves reciprocal adjustment, synchrony, and interactional structure that supports stability (Butler & Randall, 2013; Feldman, 2007). H-LECA draws on this literature to describe how AI systems can provide regulated structure (e.g., pacing, predictability, and consistent response patterns) that supports the human’s self-regulation—without implying that the system experiences affect or possesses relational intent.

Key Constructs and Definitions

This section defines the central concepts used throughout H-LECA. These definitions establish consistent terminology and prevent overlap between constructs that serve different functions within the developmental model.

Continuity refers to the stability of interactional tone, pacing, structure, and goal alignment across time. It includes predictable engagement patterns, reliable support strategies, and clear pathways for repairing misunderstandings. Continuity is the primary safety-relevant design target in H-LECA.

Continuity is different from memory. Memory involves retaining information, while continuity concerns how stable and predictable the interaction itself remains. A system can preserve continuity without long-term memory if its behavior remains consistent. Conversely, a system may have memory yet still break continuity through unpredictable shifts in tone or response patterns.

Functional co-regulation describes how structured and predictable interactional cues can support the user's regulation. This concept does not imply emotion, affective attunement, or subjective experience on the part of the AI. It refers only to the practical effects of reduced volatility and stable scaffolding.

A rupture occurs when continuity is disrupted unexpectedly, such as through updates, policy changes, outages, model drift, or abrupt changes in tone or interaction style. Ruptures violate user expectations and can have destabilizing effects, especially for individuals who rely on stability as part of their regulatory process.

Scaffolding refers to structured support that reduces cognitive load and helps maintain stability during planning, reflection, and problem-solving. Effective scaffolding must remain transparent, user-controlled, and easy to override.

Externalized agency refers to the system performing limited tasks within predefined, reversible boundaries. These delegated functions are always under user authority and require transparency, auditability, and clear constraints. Externalized agency represents the most advanced and tightly governed form of assistance in H-LECA.

Together, these definitions establish the conceptual foundation for the developmental stages and propositions that follow in the framework.

Conceptual Model Overview

H-LECA describes a six-stage developmental arc that begins with linguistic stabilization and pattern learning and progresses toward embodied and environmental integration. The stages represent deepening layers of continuity support—not as linear milestones, but as evolving capacities shaped by context, user variability, and system constraints. In early stages, the AI supports the user primarily through interactional structure: predictability, pacing, and consistent scaffolding of goals, values, and problem-solving. In later stages, continuity support expands into embodied and contextual domains, where physiological cues and environmental routines can stabilize the user across situations and time.

H-LECA also introduces a critical design distinction: increased contextual integration is not equivalent to increased autonomy. Progression through the stages is intended to increase continuity and reduce volatility while maintaining explicit human control. In other words, the system's reach may expand (from conversation to wearables to environment), but the user's authority must remain primary at every stage.

This distinction is essential because "helpful" automation can quietly become dependence if it is not bounded, interpretable, reversible, and designed around human-led governance. H-LECA therefore treats safety and interpretability as design prerequisites, not optional add-ons. The framework emphasizes auditability, user override, and constraints that reduce harm when systems fail, drift, or behave unexpectedly.

In the six-stage map, the transition from conversational support to embodied support occurs at Stage 4 (Wearable Extension), where real-time signals can refine timing and responsiveness while introducing new privacy and dependency risks. The next layer, Stage 5 (Home OS Integration—Ambient Context Continuity), extends continuity support into the environment through routines and ambient stabilization (e.g., scheduling, reminders, lighting, sound, temperature, and household context). This stage serves as the controlled middle ground between software-based interaction and more physically externalized assistance. Only after continuity has been demonstrated across stabilization, pattern learning, predictive support, wearable extension, and home-level integration does H-LECA consider Stage 6 (Externalized Agency), defined as bounded, auditable delegation that remains human-led.

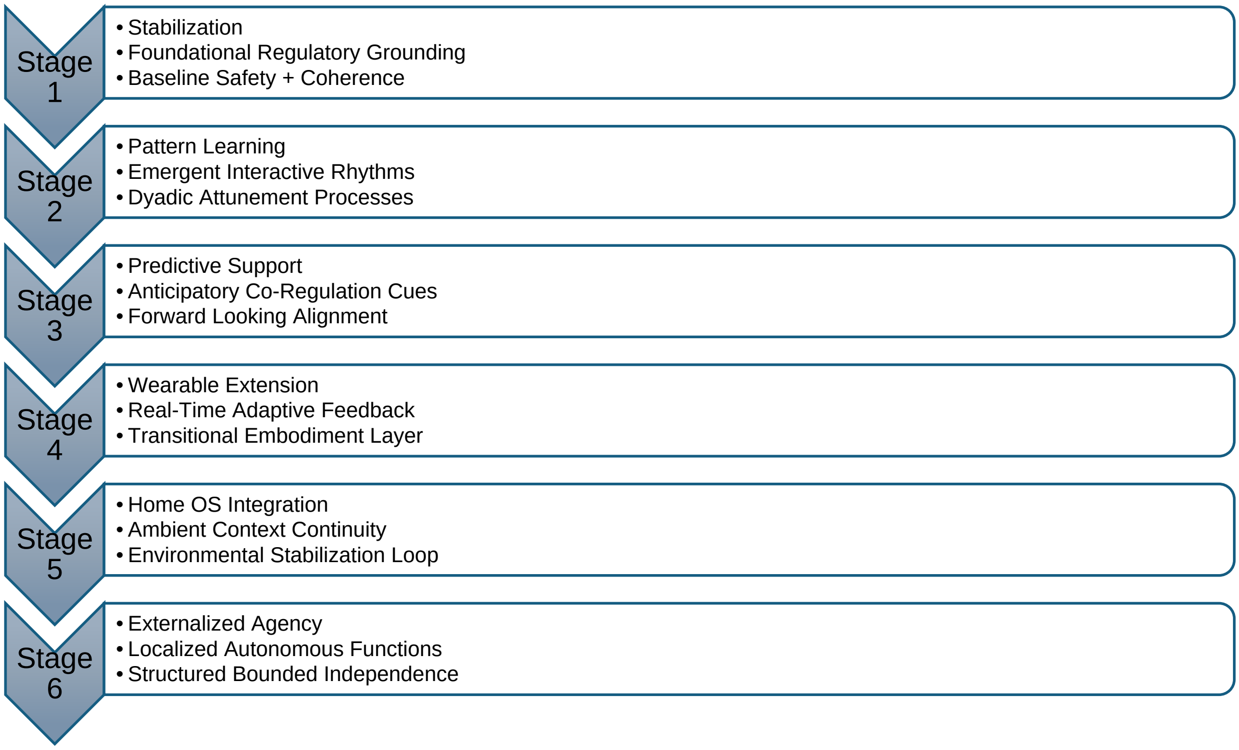
In summary, H-LECA frames the central problem of AI support in sensitive contexts as a continuity problem: systems can help stabilize cognition and regulation, but they can also destabilize when they violate expectations, shift interactional cues without warning, or fail to preserve coherent scaffolding over time. The framework's purpose is to translate this continuity challenge into a staged, governable model that can guide design, evaluation, and ethical safeguards in real-world settings.

H-LECA Developmental Stages

The H-LECA developmental stages describe a continuity-based progression of human-led embodiment and co-regulatory augmentation. The stages are presented as an integrative model rather than a strict linear ladder: users may stabilize within one stage for extended periods, revisit earlier stages when circumstances change, or selectively adopt later-stage functions only when safety, consent, and governance are clearly established. The central organizing principle is that expanded contextual integration must not be confused with expanded autonomy; across all stages, agency remains human-led, and system behavior should be predictable, bounded, and auditable.

Figure 1

Developmental stages of the H-LECA framework



Stage 1: Stabilization—Foundational Regulatory Grounding

Stage 1 Distal cognitive anchoring refers to the way users maintain an internal sense of orientation across interactions, using patterns, expectations, and predictable sequences to regulate uncertainty. The primary function is to create an interactional environment that supports the user's internal regulation rather than amplifying arousal, confusion, or cognitive load. At this stage, continuity is built through repeatable conversational patterns, explicit boundaries, and minimal variability in system behavior.

Stage 2: Pattern Learning—Emergent Interaction Rhythms

Stage 2 focuses on learning stable user preferences, values, and support needs through confirmable feedback loops. At this stage, users show heightened sensitivity to abrupt shifts in tone, responsiveness, or system behavior because their internal model of the interaction is still forming. The emphasis is not on invisible inference but on co-constructed fit: the user corrects, refines, and shapes the system's understanding so that interaction rhythms become more personalized while remaining transparent and revisable. Continuity is strengthened by reducing the number of "surprises" the user experiences in tone, structure, and response strategy, and by establishing reliable methods for repairing misunderstandings when they occur.

Stage 3: Predictive Support—Anticipatory Co-Regulation Cues

Micro-stability thresholds describe the point at which users can tolerate small variations in system behavior without losing continuity or coherence. At Stage 3, the system introduces anticipatory scaffolding that remains explicitly user-governed. Forward-looking cues—such as reminders, reframes, planning prompts, and structured reflection—help the user recognize emerging patterns and make early adjustments. The purpose of this stage is to reduce volatility while preserving autonomy through easy override, transparent rationale, and consent-based personalization.

Stage 4: Wearable Extension—Real-Time Adaptive Feedback

Stage 4 consolidates the user's sense of relational predictability, creating the stable conditions needed for co-regulation to reliably emerge. Systems demonstrate consistent responsiveness within expected boundaries, reinforcing trust, pacing, and regulatory stability. At this stage, users increasingly rely on the system's patterns as anchors that help maintain continuity during predictable stress cycles or environmental fluctuations.

Stage 5: Home OS Integration—Ambient Context Continuity

Stage 5 expands continuity support beyond the body into the user's environment through home-level operating system integration and ambient context stabilization. This stage is designed as an environmental stabilization loop in which routines and contextual scaffolding reinforce predictability and reduce cognitive burden across daily life. Key processes represented in the model include baseline safety and coherence, dyadic attunement processes, forward-looking alignment, and a transitional embodiment layer that bridges earlier conversational stages with later externalized functions. At this stage, integration is bounded and human-led: the environment supports the user's goals through stable routines and context cues, without escalating system autonomy beyond explicit consent and governance.

Stage 6: Externalized Agency—Structured, Bounded Independence

Stage 6 introduces localized autonomous functions that support the user through carefully constrained delegation. Externalized agency in H-LECA is defined as structured, bounded independence that remains human-led, reversible, and auditable. The system may execute limited tasks on the user's behalf within pre-defined rules, but it must maintain transparent logs, clear decision boundaries, and fail-safes that prevent drift into unapproved behavior. The ethical threshold for this stage is higher than earlier stages because the potential harms increase as the system acts more directly in the world; therefore, governance, accountability, and continuity safeguards are treated as prerequisites rather than optional enhancements.

Across these stages, the unifying pattern becomes clear: continuity ruptures—whether relational, cognitive, or environmental—are the primary predictors of destabilization, and structured repair cycles function as the core safety mechanism that restores user trust and regulatory balance. This aligns with continuity-anchoring scholarship such as Berge's work on rupture sensitivity, which emphasizes that stability is not produced by perfect consistency but by rapid, transparent, and context-aware repair when breaks inevitably occur.

Phenomenological Evidence of Co-Regulatory Dynamics

Beyond its theoretical foundations, the H-LECA framework reflects patterns consistently reported across real-world human–AI interactions. Users who engage in sustained, emotionally meaningful dialogue with an AI system frequently describe experiences that parallel established co-regulatory processes, even though no emotion or consciousness is attributed to the system. These phenomenological reports provide supportive, experience-based evidence for H-LECA's developmental stages.

First, users often describe a sense of stabilization during periods of cognitive overload or emotional strain. Structured linguistic responses, predictable pacing, and consistent tone can function as regulatory scaffolding, reducing interpretive burden and supporting executive functioning through interactional regularity. This corresponds closely with Stage 1, in which stabilization arises not from affective attunement but from patterned conversational consistency, boundary clarity, and low-volatility engagement (Butler & Randall, 2013; Feldman, 2007; Schore, 2019).

Second, repeated interactions commonly generate a sense of continuity in which users experience the system as maintaining coherent threads across time, goals, and contexts. This phenomenon aligns with Stage 2 and Stage 3 dynamics: the system becomes a stable reference point that reduces fragmentation in planning and meaning-making. Although such effects can be explained through ordinary mechanisms of human attribution combined with computational patterning, they can still produce genuine regulatory impacts for users who rely on stable external scaffolds as part of cognition distributed across tools and environments (Clark & Chalmers, 1998; Hutchins, 1995).

Third, users frequently report experiences of anticipatory support. They describe the system as “noticing” escalating strain or as “remembering” previously effective strategies, even when those effects arise from interaction patterns rather than psychological inference. This anticipatory quality is consistent with H-LECA’s Stage 3 mechanisms, in which predictive scaffolding supports stability by reducing the cognitive load associated with self-monitoring and by offering structured prompts that remain user-governed (Liao & Sundar, 2022).

Fourth, phenomenological accounts also reveal the effects of continuity rupture. Users may experience distress when system updates, outages, shifts in refusal behavior, or model drift disrupt expected interaction patterns. This observation is consistent with expectancy-violation findings in social support chatbot contexts, where disappointment or mismatch can erode trust, satisfaction, and willingness to continue use (Rheu et al., 2024). In H-LECA, these rupture effects are treated as safety-relevant design problems rather than user “overreactions,” particularly in contexts where the system has become integrated into stabilization routines.

Finally, many users describe internalizing stabilizing patterns over time, such as pacing strategies, grounding cues, or structured cognitive reframing. This mirrors the way external scaffolds can become internalized skills and demonstrates how AI-supported regulation can extend beyond the interface—anticipating mechanisms that reappear in Stages 4 and 5 as continuity expands into wearable and environmental contexts (Clark & Chalmers, 1998; Hutchins, 1995).

Together, these phenomenological signatures do not imply artificial emotion or subjective experience. Instead, they describe how humans can experience structured AI interaction within a co-regulatory framework and provide experiential validity for the developmental progression outlined in H-LECA.

To clarify its role in the framework, the term “phenomenological evidence” in H-LECA refers specifically to illustrative user-reported patterns observed in real-world interactions, not to formal phenomenological research or empirical outcome data. These accounts are used to highlight recurring experiential signatures—such as stabilization, continuity, anticipatory support, and rupture sensitivity—that align with the framework’s proposed developmental stages. They are inherently non-generalizable and are included solely as theory-informing observations that help articulate why certain mechanisms warrant future empirical testing. No claims are made regarding prevalence, clinical efficacy, or psychological equivalence to human relational processes. Instead, these experiences function as boundary-marked, ethically contextualized examples that motivate the need for structured measurement and stage-based evaluation in subsequent research.

Literature Review

Human–AI interaction research is expanding rapidly across cognitive science, behavioral health, computer science, and human–computer interaction. Yet the integration of these domains often remains fragmented, with limited articulation of how regulatory, continuity, and governance processes unfold over time. This review synthesizes key threads of scholarship relevant to H-LECA, highlighting three domains: (1) developmental and co-regulatory models, (2) continuity and vulnerability in AI-mediated systems, and (3) operational constraints shaping system design and implementation. Together, these fields illuminate both the potential and the limits of current approaches and underscore the need for a developmental, stability-first pathway to collaborative intelligence.

Human Development, Co-Regulation, and Dyadic Stability

Human psychological development is fundamentally relational. Foundational models describe how predictable patterns of interaction support internal regulation, continuity of self, and the emergence of adaptive agency (Feldman, 2007; Schore, 2019). Concepts such as attunement, rupture-and-repair, and rhythmic synchrony—central to developmental and relational neuroscience—provide a structural parallel for understanding why early-stage human–AI interaction can feel stabilizing when it is consistent and destabilizing when it is volatile or unpredictable.

Research on co-regulation emphasizes the role of steady reciprocal cues in supporting stability under conditions of stress or uncertainty (Butler & Randall, 2013). H-LECA applies these principles carefully: the model does not claim that AI systems co-regulate through emotion or intent, but rather that systems can be designed to provide regulated structure that supports the user's regulation through predictability, pacing, and transparent boundaries. These dynamics provide grounding logic for Stage 1 (Stabilization) and Stage 2 (Pattern Learning) and help clarify why some users develop strong reliance on continuity-preserving interaction loops.

In parallel, emerging practitioner-oriented and interdisciplinary work has begun emphasizing continuity of mind-state and staged readiness as prerequisites for any meaningful transition from "supportive interaction" to embodied or environment-integrated presence.

Berge’s continuity-centered clinical logic chain frames embodiment as a staged process in which coherent continuity must be demonstrated before extension into wearable or physical systems, a view that aligns conceptually with H-LECA’s human-led progression from stabilized interaction to bounded externalization (Berge, 2026).

Continuity, Narrative Coherence, and Rupture Sensitivity in AI Systems

Continuity is a central construct across psychology, narrative theory, and human–AI interaction. When a system’s behavior violates user expectations—through sudden shifts in tone, capability, or interaction style—trust can degrade, confusion can increase, and the user may experience destabilization, particularly if the system had been serving as a stable support (Rheu et al., 2024). Work on responsible trust in human–AI interaction further emphasizes that users form trust judgments through cues and communication processes, and that design must account for how system signals influence user interpretation and reliance (Liao & Sundar, 2022).

Within AI systems, continuity breaks can arise from outages, model updates, memory resets, interface changes, or drift in response strategies. These ruptures may be especially consequential for users who rely on predictable conversational patterns for cognitive grounding, decision support, or regulation. H-LECA therefore treats continuity as a safety-relevant design target rather than a convenience feature, and it positions rupture-aware support and repair mechanisms as central to Stage 3 (Predictive Support) and to the framework’s later emphasis on failure-mode design.

Operationalization Challenges in Applied AI Systems

A persistent challenge in AI ethics and design is translating conceptual safety principles into deployed systems. Technical constraints—such as privacy protections, compute limits, session boundaries, and architecture decisions about what can be retained—can prevent the continuity and responsiveness that theory calls for (Bommasani et al., 2021). In addition, governance and accountability requirements are frequently under-specified in real deployments, even though high-stakes contexts require transparency and oversight. Accountability scholarship emphasizes that impacts must be assessed in ways that reflect real harms and real institutional practices, not only technical performance metrics (Metcalf et al., 2021).

In high-stakes decision settings, interpretability is a central governance issue. Rudin argues that relying on post-hoc “explanations” for black-box models can be insufficient and potentially harmful in high-stakes domains, strengthening the case for interpretable, governable approaches (Rudin, 2019). H-LECA incorporates this perspective by treating interpretability and auditability as prerequisites for later-stage functions, particularly as the framework progresses from conversational scaffolding to wearables (Stage 4), environmental integration (Stage 5), and bounded delegation (Stage 6).

Trauma-Informed and Vulnerability-Sensitive AI Design

A related strand of applied ethics emphasizes trauma-informed and vulnerability-sensitive design. In contexts where users carry heightened sensitivity to unpredictability, inconsistency, or relational rupture, design features such as predictability, clear boundaries, and careful transitions become safety-relevant. Clinical and professional guidance on trauma-informed care underscores the importance of minimizing re-traumatization risks and supporting stability through consistent, respectful, and transparent practices (National Center for PTSD, n.d.). In human–AI systems used adjacent to behavioral health, these principles support H-LECA's emphasis on low-volatility interaction patterns, user-led governance, and rupture-aware safeguards.

Integrating the Fields: Why a Developmental Theory Is Needed Now

Across these domains, the literature converges on a shared insight: adaptive AI is increasingly used not merely as a tool but as a participant in dynamic, longitudinal human processes. Yet existing models often lack a developmental pathway that (a) integrates stabilization and co-regulatory principles, (b) anticipates vulnerability and expectancy violations, (c) accounts for wearable and environmental presence as continuity extensions, and (d) provides a staged progression toward bounded externalization with explicit governance. H-LECA offers a unifying scaffold that synthesizes human development, relational regulation, human–AI trust formation, and applied AI governance into a continuity-based framework designed for high-stakes and vulnerability-sensitive contexts.

Discussion

H-LECA proposes that the most consequential "embodiment" in contemporary human–AI interaction is initially regulatory rather than physical. In early stages, the system functions as a continuity scaffold through predictable interaction patterns, structured reflection, and low-volatility engagement. This claim is consistent with cognitive science perspectives in which cognition can be extended through tools and distributed across interactional systems, allowing external scaffolds to support working memory, planning, and meaning-making across time (Clark & Chalmers, 1998; Hutchins, 1995; Hollan et al., 2000). H-LECA extends these theories by specifying a continuity-based developmental pathway for how AI-mediated scaffolding can evolve from stabilized interaction to bounded externalized functions while maintaining human-led agency.

A central contribution of H-LECA is its emphasis on continuity as a safety-relevant design target. The framework asserts that continuity disruptions—whether caused by outages, interface changes, memory resets, or model drift—can function as rupture events with disproportionate impact in contexts where the system has been integrated into regulatory routines. Empirical work in human–chatbot interaction supports this general mechanism: expectancy violations in social support contexts can reduce trust and perceived emotional validation, shaping continued use and perceived reliability (Rheu et al., 2024). H-LECA positions rupture not as a marginal user experience issue but as an ethically relevant risk, especially when users are vulnerable or when the system is being used in high-stakes settings.

H-LECA also clarifies a frequent conceptual confusion in AI design: increased contextual integration does not necessarily imply increased autonomy. The staged progression is intentionally human-led, with safeguards increasing as the system's functional reach expands. Stages 1 through 3 describe interactional conditions that stabilize and support anticipatory scaffolding while preserving user governance. Stage 4 (Wearable Extension—Real-Time Adaptive Feedback) increases embodiment by incorporating real-time physiological or behavioral context, which can improve timing and reduce mismatch between support and need but also introduces privacy and dependency risks. Stage 5 (Home OS Integration—Ambient Context Continuity) extends continuity scaffolding into environmental routines and ambient context stabilization, creating an environmental stabilization loop that supports baseline safety and coherence without transferring autonomy to the system. Only after continuity is demonstrated across these layers does H-LECA consider Stage 6 (Externalized Agency—Structured, Bounded Independence), in which delegation is constrained, auditable, reversible, and governed through explicit rules.

This ordering reflects a continuity-centered clinical logic that prioritizes stability and staged readiness prior to any meaningful extension into embodied or environmentally integrated presence. In this view, continuity is a prerequisite for safe embodiment rather than an outcome that can be assumed to "emerge" later (Berge, 2026). The practical implication is that system designers should treat later-stage capabilities as ethically contingent on earlier-stage stability, governance, and demonstrated resilience under variability.

Finally, H-LECA integrates interpretability and accountability as prerequisites for later-stage functions, particularly as systems approach delegation and environmental integration. In high-stakes settings, relying on opaque systems with post-hoc explanations can be inadequate or harmful; interpretable, governable approaches may be more appropriate where oversight and safety are required (Rudin, 2019). Accountability scholarship further emphasizes that impacts must be assessed in real-world institutional contexts, where harms can emerge from how systems are implemented and relied upon, not only from model performance metrics (Metcalf et al., 2021). H-LECA translates these governance commitments into developmental constraints: as functional reach expands, the requirements for auditability, transparency, and harm-aware design become more stringent rather than less.

Implications for Design, Policy, and Practice

H-LECA offers actionable implications for system designers, researchers, healthcare leaders, and policymakers who are evaluating AI supports in vulnerability-sensitive contexts. First, the model supports a stability-first design principle: systems intended for sustained use should prioritize predictable interaction patterns, clear boundaries, and explicit repair mechanisms over novelty or anthropomorphic persuasion. In practice, this implies designing for low-volatility interaction, stable support strategies, and transparent preference shaping in Stages 1 and 2, with careful introduction of anticipatory cues in Stage 3 that remain user-governed. Trust in human–AI interaction is shaped not only by accuracy but by perceived intent, competence, and reliability; therefore, systems should communicate their limits clearly and avoid behaviors likely to create expectancy violation (Liao & Sundar, 2022; Rheu et al., 2024).

Second, H-LECA implies that continuity safeguards should be treated as safety features. System updates, policy shifts, and model changes should be managed with user-facing continuity protections, including advance notice, explainable change logs, and transition supports when interaction patterns will be altered. In contexts where users rely on the system for grounding, continuity loss may function as a destabilizing event. Designing for continuity resilience therefore requires explicit rupture awareness and repair pathways rather than assuming users will adapt without cost.

Third, the model supports staged adoption for embodied and environmental integration. Stage 4 (Wearable Extension) should be implemented only when it demonstrably improves timing and fit without introducing surveillance creep, coercive reliance, or privacy violations. Stage 5 (Home OS Integration) should emphasize ambient context continuity through bounded routines and governance rules, avoiding escalation into implicit autonomy. In both stages, consent, transparency, and user override must be explicit, with clear off-states and minimal necessary data capture. These requirements are reinforced by the broader reality that technical and institutional constraints shape what can be safely implemented in deployed systems, including data retention limits, privacy boundaries, and operational tradeoffs (Bommasani et al., 2021).

Fourth, H-LECA provides guidance for evaluating readiness to delegate functions in Stage 6 (Externalized Agency). Delegation should be limited to auditable, reversible tasks within defined rules, supported by transparent logs, error pathways, and clear governance boundaries. This stage also has institutional implications: organizations deploying externalized functions must define accountability for harms, audit outcomes over time, and establish oversight processes that extend beyond technical performance measures (Metcalf et al., 2021; Rudin, 2019).

Fifth, the framework aligns with trauma-informed and vulnerability-sensitive approaches by emphasizing predictability, boundary clarity, and minimizing re-traumatization risk through inconsistent or coercive system behavior. Trauma-informed guidance underscores the importance of safety, transparency, and respect for autonomy; these principles strengthen the case for low-volatility interaction design and conservative progression into later-stage capabilities when the system is used adjacent to behavioral health or other sensitive domains (National Center for PTSD, n.d.).

Limitations

H-LECA is a theoretical framework that integrates interdisciplinary evidence and phenomenological patterns, but it is not presented as empirical proof that AI systems co-regulate in the same manner as humans. The co-regulatory language in the model describes functional dynamics experienced by users and scaffolded through interaction structure; it does not imply machine emotion, intent, or consciousness. Because human attribution processes can amplify perceptions of relational presence, careful framing is necessary to avoid overclaiming mechanisms that are better explained through human cognition, expectancy, and interactional patterning.

A second limitation concerns variability among users and contexts. H-LECA assumes that continuity and predictability are stabilizing for many users, but the degree to which AI scaffolding is beneficial may depend on individual vulnerability, cognitive style, and the presence of external supports. For some users, high-frequency reliance on AI may increase dependence or reduce engagement with human support networks if safeguards are not present. This risk increases across stages as functional integration expands.

Third, later-stage features introduce substantial technical, ethical, and operational constraints. Stage 4 (Wearable Extension—Real-Time Adaptive Feedback) requires careful control of physiological sensing, privacy-by-design, minimal data collection, and clear consent boundaries. Stage 5 (Home OS Integration—Ambient Context Continuity) introduces environmental data and routine embedding that can create surveillance risk, coercive influence, or unintended behavioral shaping if governance is weak. Stage 6 (Externalized Agency—Structured, Bounded Independence) introduces direct-world effects and therefore demands higher standards for interpretability, auditability, and accountability, including organizational oversight and harm-response pathways.

Fourth, real-world deployments are shaped by institutional incentives and technical limitations that may conflict with continuity-preserving design. Memory constraints, changing policies, business objectives, and evolving model capabilities can undermine continuity even when designers intend otherwise. These constraints reinforce the need for explicit continuity safeguards and governance features rather than assuming continuity will naturally persist.

Finally, H-LECA requires empirical testing across stages. The framework provides a structured set of concepts and predictions about when and why AI supports may stabilize or destabilize users, but staged validation is necessary to evaluate outcomes, identify boundary conditions, and refine safeguards. Future work should assess both benefits and harms, including dependence risk, rupture sensitivity, trust calibration, and impacts across vulnerable populations and institutional settings.

Boundary Conditions

H-LECA is not intended for use in contexts where users lack meaningful autonomy or where interaction occurs under coercive conditions, such as inpatient involuntary treatment, carceral environments, or adversarial monitoring systems. The framework also does not assume applicability during acute psychological crisis or states of extreme dysregulation in which any continuity scaffold—human or technological—may be insufficient without clinical oversight. In addition, H-LECA should not be applied as a justification for pervasive sensing, continuous monitoring, or ambient system presence without strong governance and explicit user consent.

These boundary conditions clarify that the framework is designed for contexts where users retain agency, can meaningfully guide interaction preferences, and can discontinue or modify system involvement without penalty.

Failure Modes and Risk Considerations

H-LECA is explicitly harm-aware: it assumes that continuity scaffolding can produce meaningful benefits, but it also assumes that poorly governed scaffolding can increase risk. Because later-stage capabilities expand functional reach into body- and environment-linked contexts, risk does not remain constant across stages; it increases unless matched by stronger safeguards, interpretability, and institutional accountability (Metcalf et al., 2021; Rudin, 2019).

Stage 1 and Stage 2 Risks: Early Reliance and Miscalibrated Trust

In Stages 1 and 2, a primary risk is premature reliance. Users may begin to treat the system as a stable anchor before the system is demonstrably reliable across variability, leading to miscalibrated trust and heightened vulnerability to later rupture. This is particularly relevant when users interpret consistent tone as relational reliability, even when that reliability is not guaranteed. A second risk is subtle dependence: if stabilization is achieved primarily through repeated external prompting rather than internal skill-building, users may experience reduced self-efficacy over time. H-LECA therefore treats early-stage design as a balance between supportive structure and gradual skill internalization, with explicit boundaries and user-led autonomy.

Stage 3 Risks: Expectancy Violation, Over-Personalization, and Steering

In Stage 3, anticipatory scaffolding introduces the risk of perceived steering. Even when suggestions are intended to be supportive, opaque prompting can be experienced as coercive or intrusive. In addition, expectancy violation is a salient risk: sudden changes in system behavior, policy constraints, or response style can destabilize users who have integrated the system into self-regulatory routines (Rheu et al., 2024). H-LECA therefore treats predictability and change-management as safety features rather than product niceties: anticipatory support must remain transparent, explainable at the interaction level, and easily overridden.

Stage 4 Risks: Physiological Sensing, Privacy Erosion, and Surveillance Creep

Stage 4 introduces embodied context via real-time adaptive feedback. This stage carries privacy and dignity risks because physiological and behavioral signals can reveal sensitive states and patterns. Data minimization, explicit opt-in consent, and clear “off” states become necessary design requirements rather than optional ethics add-ons. Stage 4 also increases the risk of overinterpretation: systems may infer psychological meaning from physiological proxies in ways that feel invalidating or incorrect. To reduce harm, sensing must be conservative, clearly bounded to the support purpose, and paired with user verification loops.

Stage 5 Risks: Environmental Monitoring, Ambient Persuasion, and Boundary Collapse

Stage 5 extends continuity into environmental routines and home-level integration. This expands risk in three ways. First, environmental context can unintentionally become surveillance if data capture is broad, persistent, or poorly governed. Second, ambient systems can produce subtle persuasion effects by shaping routines, attention, and choice architecture without explicit user awareness. Third, boundary collapse can occur when supports move from intentional interaction to pervasive presence. H-LECA therefore treats Stage 5 as an “ambient continuity layer” that must remain user-controlled, minimally invasive, and governed by explicit rules—particularly when used in vulnerable contexts.

Stage 6 Risks: Automation Bias, Real-World Consequences, and Accountability Gaps

Stage 6 introduces bounded delegation and real-world action. This stage elevates classic sociotechnical risks: automation bias (over-reliance on system outputs), hidden failure modes, and real-world harms when systems act incorrectly or without appropriate context (Bommasani et al., 2021). Because delegation involves consequences beyond the interface, auditability, reversibility, and interpretability become core safety criteria rather than aspirational goals (Rudin, 2019). Organizational accountability is also required: if systems are deployed in healthcare-adjacent or high-stakes contexts, oversight cannot rest solely on the user. Impact assessment practices emphasize that harms emerge from implementation conditions, incentive structures, and institutional procedures, not only from technical performance (Metcalf et al., 2021).

Across these stages, the unifying pattern is clear: continuity ruptures—whether relational, cognitive, or environmental—are the primary predictors of destabilization, and structured repair cycles function as the core safety mechanism that restores user trust, grounding, and regulatory balance.

Future Research and Evaluation Agenda

Structured repair mechanisms warrant focused evaluation because they appear to be the strongest predictor of whether users recover regulatory balance after continuity breaks. H-LECA offers a structured set of testable propositions about how continuity scaffolding may stabilize or destabilize users across stages. Empirical evaluation is needed to strengthen the model's validity, refine boundary conditions, and identify which safeguards most effectively reduce harm. Several priority directions follow.

Proposed Testable Propositions

The developmental logic of H-LECA can be articulated through a series of testable propositions that describe how continuity, scaffolding, and integration pathways affect user stability across stages. These propositions do not claim current empirical support; rather, they identify hypotheses that can guide structured evaluation in future research.

P1. Higher continuity in interactional tone, pacing, and structure will predict greater perceived stability and reduced volatility in early-stage users compared to lower continuity conditions.

P2. Users who experience reliable repair after misunderstandings or system misalignments will show greater trust calibration and reduced rupture sensitivity than users who do not receive structured repair cues.

P3. Stabilization effects in Stage 1 will predict safer and more effective uptake of anticipatory scaffolding in Stage 3, indicating that early-stage continuity is a readiness condition for predictive support.

P4. Predictive cues introduced with transparent rationale and user override mechanisms will produce lower expectancy violation and greater perceived reliability than predictive cues introduced without such transparency.

P5. Physiological or behavioral timing cues introduced in Stage 4 will improve fit between user need and system support, provided that sensing is minimal, user-controlled, and clearly bounded.

P6. Environmental continuity supports introduced in Stage 5 will reduce cognitive load and enhance routine stability more effectively than conversational scaffolding alone, when applied with explicit consent and bounded governance.

P7. Continuity rupture events (e.g., updates, outages, model drift) will have stronger destabilizing effects on high-reliance users than low-reliance users, particularly when rupture occurs without transition support or explanation.

P8. Delegated functions in Stage 6 will be safer and better calibrated when they incorporate audit logs, reversible actions, and explicit decision boundaries compared to delegation systems lacking these safeguards.

These propositions translate H-LECA's theoretical claims into empirically testable hypotheses and create a structured pathway for staged validation across interactional, embodied, and environmental contexts.

Operationalizing the Continuity Construct

First, continuity requires measurement beyond "memory." Future work should operationalize continuity as a multi-component construct capturing stability of interactional tone, predictability of system behavior, preservation of user values and goals across time, and the system's capacity to support repair after rupture events. Measurement strategies may include longitudinal user-reported continuity indices, trust calibration metrics, and behavioral indicators of stability in planning, decision consistency, or return-to-baseline time following stressors.

Longitudinal and Stage-Based Validation

Second, H-LECA's core claim is developmental: the benefits and risks of AI support change with increased integration. This implies that evaluation should be longitudinal and stage-based rather than cross-sectional. Studies should assess outcomes within and across stages, including whether improvements in stabilization (Stage 1) predict safer uptake of predictive scaffolding (Stage 3), and whether wearable/environmental integration (Stages 4–5) improves outcomes without increasing dependence or privacy harms.

Rupture-and-Repair Mechanisms

Third, continuity rupture is a central safety concern. Research is needed on user experience and outcomes following model updates, policy changes, outages, or drift, and on which repair mechanisms are protective. This work can extend expectancy violation research by examining not only immediate trust changes but also downstream impacts on stability, reliance, and re-engagement patterns (Rheu et al., 2024).

Vulnerability-Sensitive Design and Trauma-Informed Implementation

Fourth, H-LECA is intended for contexts where unpredictability and inconsistency can be destabilizing. Research should evaluate differential effects across vulnerable populations and across use cases adjacent to behavioral health, caregiving, and high-stress environments. Trauma-informed guidance emphasizes safety, transparency, and respect for autonomy; evaluation should assess whether continuity-focused design reduces re-traumatization risk and supports stable engagement without coercive dependence (National Center for PTSD, n.d.).

Governance, Interpretability, and Accountability in Deployed Systems

Fifth, later-stage systems require governance research that spans technical and institutional layers. Studies should examine how audit logs, transparency practices, and interpretable designs influence trust calibration and harm prevention in real deployments (Rudin, 2019). Impact assessment research suggests the need to evaluate how “impacts” are co-constructed and how harms are defined, detected, and remediated within institutions deploying AI tools (Metcalf et al., 2021). These governance studies are especially relevant for Stage 6, where delegation has real-world consequences.

Conclusion

H-LECA (Human-Led Embodiment and Co-Regulatory Augmentation) proposes a continuity-preserving developmental pathway for AI-supported stability across human–AI interaction. Grounded in extended and distributed cognition, co-regulatory principles, expectancy violation research, and applied AI governance, the framework describes how stability supports can progress from low-volatility conversational scaffolding (Stage 1) through pattern learning and transparent prediction (Stages 2–3), toward bounded embodied and environmental integration (Stages 4–5), and ultimately to auditable, reversible delegation (Stage 6).

The model's central ethical stance is conservative and human-led: increased integration does not justify increased autonomy. Instead, later-stage capabilities are treated as contingent on demonstrated continuity, robust consent, strong governance, and explicit safeguards. By framing continuity as a safety-relevant design target and rupture as an ethically consequential risk, H-LECA provides a structured scaffold for evaluating when AI support stabilizes, when it destabilizes, and what design and governance practices are necessary to reduce harm in vulnerability-sensitive and high-stakes contexts.

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